D. Rae Taylor and Trees (easy version)

time limit per test 3 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, data structures, dp, dsu, greedy, implementation, trees |
https://codeforces.com/problemset/problem/2171/D

"To think a commoner would even fathom sitting next to me. Know your place!"

— Claire François

This is the easy version of the problem. The only difference between the easy and hard versions is that the hard version asks you to construct an example of a satisfactory tree.

As an Earth mage, Rae has mastered the spell of growing trees! But Manaria brags that she can grow a more impressive species of trees. Rae remembers that the most rare type of tree can be grown using a formula represented by a certain permutation — please help her construct it!

You are given a permutation[] p of length n.

Determine if there exists an undirected tree with n vertices labeled 1, 2, …, n, satisfying the following condition:

- Let u and v ($1 \le$ color*redu* lt $v \le n$) be any two vertices connected by an edge. Then u appears before v in p.

[]A permutation of length n is an array that contains every integer from 1 to n exactly once, in any order.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($2 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers, $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$). It is guaranteed that all p_i are distinct.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output on a single line "`Yes`" if there exists a tree satisfying the given condition, and "`No`" otherwise.

You may output the answer in any case (upper or lower). For example, the strings "`yEs`", "`yes`", "YES", and "`yeS`" will be recognized as "`Yes`".

Example

Input

```
9
6
1 3 4 5 2 6
4
3 4 1 2
5
4 3 5 1 2
4
1 2 3 4
7
4 3 5 7 6 2 1
6
2 4 6 1 3 5
3
2 1 3
4
2 4 1 3
6
4 2 6 5 1 3
```

Output

```
Yes
No
No
Yes
No
Yes
Yes
Yes
Yes
```

Note

In the first example, we can construct the tree with the following edges:

- {3, 1},
- {4, 1},
- {6, 5},
- {6, 2},
- {6, 1}.

Then we have that

- 1 lt 3, and 1 appears before 3 in *p*,
- 1 lt 4, and 1 appears before 4 in *p*,
- 5 lt 6, and 5 appears before 6 in *p*,
- 2 lt 6, and 2 appears before 6 in *p*,
- 1 lt 6, and 1 appears before 6 in *p*.

In the second example, it can be shown that there does not exist a tree satisfying the given constraints.